

Part 2

Random Variables, Probability Distributions

Binomial Distribution

Normal Distribution, Standard Normal Distribution

The Central Limit Theorem

Poisson Distribution

Population versus Sample

Population – The set of all objects under consideration. Objects in a population are called *elements*.

Examples

- Boxes of a particular brand of corn flakes.
- Newly made light bulbs in a warehouse at a given time.

Another Definition of Population

- All possible outcomes of an *experiment* can also be values in a population.
- Flipping a coin twice. Possible outcomes are: HH, HT, TH, TT. Could make the outcomes numeric using 3, 2, 1, 0 to be the elements corresponding to the above *events*.
- If the experiment is repeated many times, the resulting values constitute a population: 2, 0, 3, 1, 0, 2, ...

Random Sampling

- **Sample** – A subset of the elements from a population.
- **Simple Random Sample** of size n from a **finite population** is a subset selected in such a way that every subset of n elements is equally likely to be the one selected. (Abbreviation: SRS)
- A random sample from a small finite population can be obtained using the **Random Numbers** - using code or random number table.
 - `sample(1:100,3)` or `sample(3)`
- Computer programs are used to draw samples from large populations.

Probability, Random Variables, and Probability Distributions

- Assume a numeric finite population of N elements, and suppose n_1 of the elements are the number k .
- Suppose a SRS of size 1 is taken from this population.
- The **probability** that the number k is selected is n_1/N .
- That is, we define this probability to be the relative frequency of occurrence of k in the population.

Probability: Example

- Example: $N = 10$
- Population: 1, 1, 1, 3, 7, 7, 9, 9, 9, 9
- Denote by $P(k)$, the probability of the number k being selected
- $P(1) = 3/10$, $P(3) = 1/10$, $P(7) = 2/10$,
 $P(9) = 4/10$, $P(2) = 0/10$, $P(1 \text{ or } 3) = 4/10$,
 $P(\text{not } 7) = 1 - 2/10$.

Random Variables

A **random variable** is a function defined on a numeric population. We will often say that this function value is **the value assumed by the random variable**.

- **Example:** $N = 10$
- **Population:** 1, 1, 1, 3, 7, 7, 9, 9, 9, 9
- Let X be a random variable defined on this population.
- X can assume any of the values 1, 3, 7, or 9.

Random Variables (continued)

- A **Random variables** are different from ordinary variables because they assume values from the population with associated probabilities.
- For e.g., we will say
$$P(X = 1) = 3/10, \quad P(X = 7) = 2/10,$$
$$P(X = 2) = 0/10, \quad P(X = -6) = 0/10,$$
$$P(1 \leq X \leq 3) = 4/10, \quad P(1 \leq X \leq 9) = 1. \text{ etc.}$$
- A **probability distribution** specifies the probabilities associated with all possible values the random variable X can assume.

The M & M Example

Example:

If X represents the number of RED *M&Ms* in a vending machine size package, then perhaps the probability distribution of X is something like the following:

x	5	6	7	8	9	10	11
$P(X = x)$	$\frac{2}{28}$	$\frac{2}{28}$	$\frac{5}{28}$	$\frac{7}{28}$	$\frac{5}{28}$	$\frac{4}{28}$	$\frac{3}{28}$

The M & M Example (continued)

We read this table by finding a value of X , say 8, then reading the probability below it:

- The probability of observing 8 M&Ms in a package is $\frac{7}{28}$, meaning, if we open many many packages of M&Ms, about 7 out of 28 will have exactly 8 M&Ms within.

This is an example of a **discrete random variable**.

Discrete Random Variables

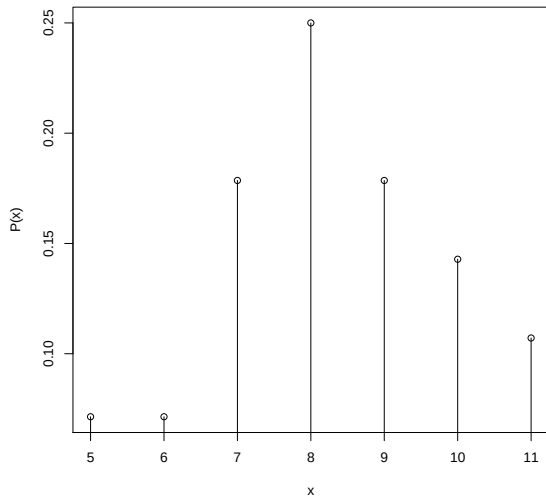
- When the elements of a population consists of a countable discrete set of values the random variable defined on that population is called **discrete random variable**.
- For discrete random variables the probability distribution can be displayed using a histogram or a table of values and associated probabilities as shown above.

Properties of Probability Distributions of Discrete Random Variable X

- $0 \leq P(X = x) \leq 1$ for all x
- $\sum_x P(X = x) = 1$

The M & M Probability Distribution

Probability Distribution for M & M Data



Continuous Random Variables

- When the population consists of an uncountable number of values in a line interval use integration to calculate proportions of the population that exist in any given **subinterval** of a continuum.
- For a random variable defined on such a population we will deal with the probability that the random variable assumes a value in a given interval, not that it assumes a specific numeric value.

Continuous Random Variables (contd.)

- When the population consists of a set of values in a continuous range the random variable defined on that population is called **continuous random variable**.
- The probability distribution of a continuous random variable is a smooth curve.
- The **area under the curve** between specified values of the random variable correspond to a probability.
- The probability that a continuous random variable assumes a specific numeric value is zero.

Continuous Random Variables (contd.)

- **Example:** A Continuous Random Variable X defined on
- a **Population:** all numbers in the interval $(0,2)$ occurring with various frequencies (that we won't specify here).
- By Definition
$$P(1/2 \leq X \leq 1/2) \equiv P(X = 1/2) = 0$$
$$P(0 \leq X \leq 2) = 1, \quad P(X < 0) = P(X \leq 0) = 0$$
- We could possibly calculate the following probability
$$P(1 \leq X \leq 2) = ?$$

Binomial Random Variable

- For some interesting populations that occur naturally, theoretical probability distributions and associated random variables, have been defined.
- One such population features elements that are only one of two kinds (Yes or No; Good or Defective; Below or Above;, etc.).
- Let us use **Success(S)** and **Failure(F)** to designate population elements, in general
- Consider a typical population of this kind consisting of m **S**'s and $N - m$ **F**'s, where N is the total number of elements in the population.

Binomial Experiment

- A **binomial experiment** consists of taking n successive SRS, each of size 1, replacing elements between draws, and counting the number of **S**'s obtained.
- The binomial experiment is conducted by sampling from the population of **S**'s and **F**'s.
- The proportion of **S**'s in the population is $\pi = m/N$ so the proportion of **F**'s is $1 - \pi \equiv (N - m)/N$.
- Let X_i be the random variable that assumes the value of the outcome of the i -th draw for each $i = 1, 2, \dots, n$.

Binomial Distribution

- Let the random variable X_i assume the value 1 if the outcome of the i -th draw is **S** or the value 0 if the outcome is **F**.
- Thus we imagine n random variables, $X_1, X_2, X_3, \dots, X_n$ each assuming a value **S** with probability π , or **F** with probability $1 - \pi$.
- That is, the probability distribution of *each* of the n random variables $X_1, X_2, X_3, \dots, X_n$ is given by:

x	0	1
$P(X = x)$	$1 - \pi$	π

Binomial Distribution (contd.)

- Now define a new random variable Y so that

$$Y = X_1 + X_2 + X_3 + \cdots + X_n$$

- Y represents the number of **Success(S)**'s in the set of values assumed by $X_1, X_2, \dots, X_n$.
- This is because each $X_i = 1$ when an **S** is drawn, and $X_i = 0$ otherwise, we can see that Y represents the total number of **S**'s in the experiment.
- Y is said to be a **binomial random variable**.

Binomial Distribution (contd.)

- That is, a **binomial random variable** assumes the value of the total number of successful outcomes in a binomial experiment.
- The random variable Y can assume any one of the values $0, 1, 2, \dots, n$ with a probability that will depend on n , the number of samples taken and π .
- Y is said to be the Binomial (n, π) random variable.
- A population on which Y is defined consists of the unique elements $0, 1, 2, \dots, n$.
- What is the proportion of each element in the population? i.e., What is the probability that Y assumes each value?

Binomial Distribution (contd.)

Theoretically, for any finite n , the probability that a **binomial random variable** Y takes on values $0, 1, 2, \dots, n$ is given by

$$P(Y = y) = \frac{n!}{y!(n-y)!} \pi^y (1-\pi)^{n-y},$$

where

$y = 0, 1, 2, \dots, n$ (number of successes in n trials)

$\pi =$ probability of success in a single trial

$1 - \pi =$ probability of failure in a single trial

$n! = n(n-1)(n-2) \dots (3)(2)(1)$

Binomial Distribution (contd.)

- In applications, we won't know π and our objective will usually be to sample the population in order to **estimate** π , and get an idea about the **error** in that estimate.
- What is the **mean** μ and **variance** σ^2 (or the standard deviation σ of a binomial random variable?
- In general, for a **discrete random variable**, these are defined as follows.

Computing the Mean and Variance of a Discrete Random Variable

- The **mean** of a discrete random variable X is:

$$E(X) = \mu = \sum_x x \cdot P(X = x)$$

- The **variance** of a discrete random variable X is:

$$V(X) = \sigma^2 = \sum_x (x - E(X))^2 \cdot P(X = x)$$

- The **standard deviation** of a random variable X is

$$\sqrt{V(X)} = \sigma$$

Computing the Mean and Variance of a Binomial Random Variable

- Using the above formula, the **mean** of a Binomial random variable Y , is shown to be $E(Y) = \mu = n\pi$
- The **variance** is shown to be $V(Y) = \sigma^2 = n\pi(1 - \pi)$.
- Thus the **standard deviation** of a Binomial random variable is $\sigma = \sqrt{n\pi(1 - \pi)}$

Binomial Distribution: Example

Consider a Binomial population with $n = 3$, $\pi = \frac{1}{3}$,
i.e., $Y \sim \text{Bin}(3, \frac{1}{3})$, The corresponding Binomial distribution is

$$P(0) = \frac{3!}{0!3!} \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^3 = 8/27$$

$$P(1) = \frac{3!}{1!2!} \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^2 = 3 \cdot \left(\frac{1}{3}\right) \left(\frac{4}{9}\right) = 12/27$$

$$P(2) = \frac{3!}{2!1!} \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right) = 3 \cdot \left(\frac{1}{9}\right) \left(\frac{2}{3}\right) = 6/27$$

$$P(3) = 1 - P(0) - P(1) - P(2) = 1/27$$

Binomial Distribution: Example (contd.)

The mean and variance of a $\text{Bin}(3, \frac{1}{3})$ population is:

- Mean = $n\pi = 3 \times \frac{1}{3} = 1$
- Variance = $n\pi(1 - \pi) = 3 \times \frac{1}{3} \times \frac{2}{3} = \frac{2}{3}$
- Standard Deviation = $\sqrt{.66667} = .8165$

Using the Binomial Distribution

- So how can this Binomial probability distribution benefit us?
- When sampling situation fits the requirements of a Binomial experiment, (at least approximately) we can view the results we obtain as the values assumed by a binomial random variable.
- This permits us to find probabilities of various possible outcomes, and use these and other properties of a binomial random variable to make inferences.
- To assume the **Binomial Model**, we need to determine whether the values of the population elements are obtained as a result of a **Binomial Experiment**.

The Binomial Model

- 1 n identical trials are made.
- 2 Each trial results in either **Success(S)** or **Failure(F)**. (only two possible outcomes).
- 3 The probability of success (π) remains the same from trial to trial.
- 4 The trials are independent.
- 5 Interest is in the number of successes obtained in the n trials. The number is viewed as being the value assumed by a Binomial (n, π) random variable Y .

The Binomial Model

- Requirement Number 3 in the definition of a binomial experiment is usually the most difficult to satisfy in practical applications.
- As trials are made, if selected elements are not replaced in the sampled population, π changes in value.
- If the population is very large, the change is very small and can be ignored.

Binomial Model: Are requirements satisfied? - Example 1

An article in the March 5, 1998, issue of The New England Journal of Medicine discussed a large outbreak of tuberculosis. One person, called the index patient, was diagnosed with tuberculosis in 1995. The 232 co-workers of the index patient were given a tuberculin screening test. The number of co-workers recording a positive reading on the test was the random variable of interest. Did this study satisfy the properties of a binomial experiment?

Binomial Model: Are requirements satisfied? - Example 1

- ① n identical trials? Yes, $n = 232$ workers
- ② Population of only S 's and F 's? Yes.
- ③ Is probability of success same in each trial? Yes if we assume the co-workers had the same risk factors as the index patient.
- ④ Independent trials? Yes, the outcome of one screening was unaffected by the outcome of the other tests.
- ⑤ Was the random variable of interest to the experimenter the number of successes y in the 232 screening tests? Yes, the number of co-workers who received a positive result.

Binomial Model: Are requirements satisfied? - Example 2

An economist interviews 75 students in a class of 100 to estimate the proportion of student who expect to obtain a C or better in the course. Is this a binomial experiment?

Binomial Model: Are requirements satisfied? - Example 2

- Population of only S 's and F 's? Yes.
- Is probability of success same in each trial? This may change from trial to trial as we may not assume sampling with replacement.

Binomial Example

A labor union's examining board for the selection of apprentices has a record of admitting 70% of all applicants who satisfy a basic set of requirements. Five members of a minority group, all satisfying the requirements, recently came before the board, and four of the five were rejected. Find the probability that 1 or fewer would be accepted if the admission rate is really 0.70.

Binomial Example - cont.

- Board decisions are viewed as random draws from a population consisting of 70% Accepts (S) and 30% Rejects (F).
- We can assume a large population so that the probability of being accepted remains constant across draws.
- Then the Binomial $(5, 0.7)$ distribution is applicable to the number of Accepts in $n = 5$ trials with $\pi = 0.7$.

Binomial Example - cont.

- Let X be the random variable representing the number of Accepts, and assume that X has a Binomial (5, 0.7) distribution. Then $P(X=0) + P(X=1) = \binom{5}{0} (0.7)^0(0.3)^5 + \binom{5}{1} (0.7)(0.3)^4 = 0.03078$
- The probability that the board will accept fewer than 2 applicants from a group of 5 is 0.03078 (which is a small probability) if indeed the true acceptance rate is 0.7.

Binomial Example - cont.

- But in fact the board DID accept fewer than 2 of the group of 5
so
- **Conclusion:** Either the board was operating to admit at its historical 70% rate and we have simply witnessed an extremely rare event, **or**
- the board has changed policy and is currently admitting at a rate less than 70%.
- One would tend to favor the latter conclusion.

Normal Random Variable

- The **Normal probability distribution** is an example of a continuous distribution.
- The associated random variable is defined in the interval $(-\infty, \infty)$.
- To model the probability distribution of many populations we often use the Normal distribution.
- Most population **distributions** are bell-shaped and can be modeled by the Normal distribution.
- We say that the associated **random variable** X , say,

$$X \sim N(\mu, \sigma^2).$$

Here μ and σ , are **parameters** of the population.

Properties of the Normal Distribution

- Probability function (or curve) of a Normal random variable X is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

where

- $E(X) = \mu$ = mean : measures the center of the distribution.
- $V(X) = \sigma^2$ = variance : measures the spread.
- $\sqrt{V(X)} = \sigma$ = standard deviation : measures the spread in same units as the mean.
- $\pi = 3.1415\dots$ $e = 2.71828\dots$

Properties of the Normal Distribution (contd.)

- The Normal distribution is symmetric about the mean μ .
- There are infinitely many Normal curves, one for each pair of values of μ and σ .
- We need to compute the areas under the curve for many of these for statistical inference.
- Mathematically speaking, proportions of a Normal(μ, σ^2) population which fall in various intervals (say (a, b) is one such interval) are obtained by evaluating the following integral (for specified values of μ, σ^2, a and b).

$$P(a \leq X \leq b) = \frac{1}{\sqrt{2\pi}\sigma} \int_a^b e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

The Standard Normal Distribution

- Table 1 lists the computed areas for the Normal distribution with $\mu = 0$, $\sigma = 1$.
- We call this the **standard normal distribution** and denote it by $N(0, 1)$.
- The relationship between a standard normal random variable Z and a normal random variable $X \sim N(\mu, \sigma^2)$ is

$$Z = \frac{X - \mu}{\sigma}.$$

The Standard Normal Distribution (contd.)

- Thus (since $\sigma > 0$) for any pair of numbers $a \leq b$,

$$\begin{aligned}P(a \leq X \leq b) &= P\left(\frac{a - \mu}{\sigma} \leq \frac{X - \mu}{\sigma} \leq \frac{b - \mu}{\sigma}\right) \\&= P\left(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}\right)\end{aligned}$$

- Thus we can compute probabilities for X for any member of the normal family using the probabilities associated with Z which are available from the Standard Normal Probabilities table.

The Standard Normal Distribution (contd.)

- The p quantiles of a Normal (μ, σ^2) population are real numbers (use x_p to denote the p quantile) such that

$$p = \int_{-\infty}^{x_p} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

- If you can find a value x that satisfies the above equation then that value will be the p^{th} quantile of the $N(\mu, \sigma^2)$ population i.e., $x = Q(p)$.
- The **standard normal quantiles** are the z -values found from the Standard Normal Probabilities table for specified p values that are in the body of the table.

Examples: Using the Standard Normal Probabilities table

- ① $P(Z \geq 0) = P(Z \leq 0) = 0.50$
- ② $P(0 \leq Z \leq 0.12) \equiv P(0 < Z < 0.12)$ =
 $P(Z < 0.12) - P(Z < 0) = 0.5478 - 0.5 = 0.0478$
- ③ $P(-0.12 \leq Z \leq 0) = P(Z < 0) - P(Z < -0.12) =$
 $0.5 - .4522 = 0.0478$
- ④ $P(-0.12 \leq Z \leq 0.12) = 2 \cdot P(0 \leq Z \leq 0.12) = 0.0956$
- ⑤ $P(0 \leq Z \leq 3) = 0.9987 - 0.5 = 0.4987$
- ⑥ $P(1 \leq Z \leq 3) = P(Z \leq 3) - P(Z \leq 1) =$
 $0.9987 - 0.8413 \qquad \qquad \qquad = 0.1574$

Examples: Using the Standard Normal Probabilities table

- 1 Find z such that $P(0 \leq Z \leq z) = 0.4846$ Equivalent to finding z such that $P(Z \leq z) = 0.9846$ which gives $z = 2.16$
- 2 Find z such that $P(z \leq Z \leq 0) = 0.2257$
Equivalent to finding z such that $P(Z \leq z) = 0.2743$ which gives $z = -0.60$
- 3 Find z such that $P(Z \geq z) = 0.011$
Equivalent to finding z such that $P(Z \leq z) = 0.9890$ which gives $z = 2.29$

Examples for $X \sim N(\mu, \sigma^2)$

- When $\mu = 1$, $\sigma = 2$, calculate $P(0 \leq X \leq 1) =$
 $P(\frac{0-1}{2} \leq Z \leq \frac{1-1}{2}) = P(-0.5 \leq Z \leq 0) =$
 $0.5 - 0.3085 = 0.1915$
- When $\mu = 3$, $\sigma = 4$, calculate $P(2 \leq X \leq 4) =$
 $P(\frac{2-3}{4} \leq Z \leq \frac{4-3}{4}) = P(-0.25 \leq Z \leq 0.25)$
 $= 0.5987 - 0.4013 = 0.1974$
- Find the 0.25 quantile of the normal population having $\mu = 3$,
 $\sigma = 4$. Find x so that $P(X \leq x) = 0.25$, or $P(Z \leq \frac{x-3}{4}) = 0.25$
 $\frac{x-3}{4} \doteq -0.67 \Rightarrow x \doteq 3 + 4 \times (-0.67) = 0.32$

Sampling Distribution of Sample Mean $\bar{Y}$.

- The outcome of drawing a random sample of size n can be characterized as the value realized by each of n random variables $Y_1, Y_2, \dots, Y_n$, each defined independently on the population.
- Thus the Sample Mean

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$$

(or any other function of $Y_1, Y_2, \dots, Y_n$) is a random variable.

Sampling Distribution of Sample Mean $\bar{Y}$.

- The probability distribution of $\bar{Y}$ is said to be the **Sampling Distribution** of $\bar{Y}$.
- It is the **derived population** of means of all possible samples of size n taken from the sampled population.
- Properties of $\bar{Y}$ include $E(\bar{Y}) = \mu$, and $V(\bar{Y}) = \sigma^2/n$, where μ and σ^2 are the population mean and variance of the original sampled population.
- The standard deviation of $\bar{Y}$ is called **Standard Error of the Mean** and is written $\sigma_{\bar{Y}}$.

The Central Limit Theorem

- The following theorem shows why the normal family of distributions is so important.
- Assume sampling is from some finite or infinite population with parameters (μ, σ^2) . Let $Y_1, Y_2, \dots, Y_n$ denote the random variables corresponding to a sample of size n
- Let $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$.
- Central Limit Theorem (CLT)

When sample size n is large, the random variable $\bar{Y}$ is approximately normally distributed with mean μ and variance σ^2/n . (i.e., $\bar{Y} \approx N(\mu, \sigma^2/n)$)

Using the Central Limit Theorem

- The approximation can be very close for n as small as 10 if the sampled population elements are symmetrically distributed around μ .
- For mildly skewed sampled populations, $n = 30$ can give a close approximation. For higher skewed populations larger n will be needed for a better approximation.
- So, the CLT says that regardless of the distribution of the sampled population, the derived population of the sample means is approximately a normal population and the approximation gets better as n gets larger.

Central Limit Theorem: Example

Estimating average weight of sheep in a large herd. Random sample of size $n = 50$. Assume $\sigma^2 = 4$. but μ , population mean animal weight in the herd is unknown.

- a) What is the approximate probability distribution of the sample mean random variable?
- b) If $\mu = 40$ what is the probability that the sample mean will be within 0.1 pounds of μ ?

Answer:

- a) By the CLT, $\bar{Y} \approx N(\mu, 4/50)$

Example continued...

b)

$$P(39.9 \leq \bar{Y} \leq 40.1) \doteq$$

$$P\left(\frac{39.9 - 40}{\sqrt{4/50}} \leq z \leq \frac{40.1 - 40}{\sqrt{4/50}}\right)$$

$$= P\left(\frac{-\sqrt{50}}{20} \leq z \leq \frac{\sqrt{50}}{20}\right)$$

$$= P(-0.35 \leq z \leq 0.35)$$

$$= 0.6368 - 0.3632 = 0.2736$$

Exercise

A random sample of 16 measurements is drawn from a population with a mean of 60 and a standard deviation of 5. Describe the sampling distribution of $\bar{y}$, the sample mean. Within what interval, symmetric about 60, would you expect $\bar{y}$ to lie approximately 95% of the time?

Solution to Exercise:

- $\bar{Y} \approx N(60, \frac{25}{16})$
- $Z = \frac{\bar{Y}-60}{5/4} \approx N(0, 1)$
- We know $P(-z_{\frac{0.05}{2}} \leq Z \leq z_{\frac{0.05}{2}}) = 0.95$ and $z_{0.025} = 1.96$ from the Standard Normal Probabilities Table

Solution to Exercise (cont.)

- Thus $P(-1.96 \leq Z \leq 1.96) = .95$ which implies that a value of Z is **expected** to lie in the interval $[-1.96, 1.96]$ approximately 95% of the time
- $-1.96 \leq \frac{\bar{y}-60}{5/4} \leq 1.96$

$$60 - 1.96 \times \frac{5}{4} \leq \bar{y} \leq 60 + 1.96 \times \frac{5}{4} \text{ giving} \\ 57.55 \leq \bar{y} \leq 62.45$$

Thus $[57.55, 62.45]$ is the interval in which $\bar{y}$, the is **expected** to lie approximately 95% of the time

Exercise 2

Psychomotor retardation scores for a large group of manic-depressive patients were found to be approximately normal with a mean of 930 and a standard deviation of 130.

- a) What fraction of the patients scored between 800 and 1,100?
- b) Less than 800?
- c) Greater than 1200?
- d) Find the 90th percentile for the distribution of manic-depressive scores.
- e) Find the interquartile range.

Solution to Exercise 2:

d.) $P(Z < a) = 0.90$ From the Standard Normal Probabilities Table,
 $a = 1.28$

Set $\frac{y-930}{130} = 1.28$ and solving for y gives $y = 930 + 166.4$
Thus $= 1096.4$ is the 90th percentile; or the 0.9 quantile.

$$P(Z < a) = 0.75$$

gives $a = 0.675$

$$P(Z < a) = 0.25$$

gives $a = -0.675$

e.) Set $\frac{y-930}{130} = 0.675$

Set $\frac{y-930}{130} = -0.675$

$$\text{Thus } Q(0.75) = 930. + 87.75$$
$$= 1017.25$$

$$\text{Thus } Q(0.25) = 930 - 87.75$$
$$= 842.25$$

$$\text{Thus } \text{IQR} = 1017.25 - 842.25 = 175.00$$

Another Exercise

The breaking strengths for 1-foot-square samples of a particular synthetic fabric are approximately normally distributed with a mean of 2,250 pounds per square inch (psi) and a standard deviation of 10.2 psi.

- a. Find the probability of selecting a 1-foot-square sample of material at random that on testing would have a breaking strength in excess of 2,265 psi.
- b. Describe the sampling distribution for $\bar{y}$ based on random samples of 15 one-foot sections.

Another Exercise - Solution

a) $Y \sim N(2250, (10.2)^2)$

$$\begin{aligned} P(Y > 2265) &= P\left(Z > \frac{2265 - 2250}{10.2}\right) \\ &= P(Z > 1.47) = 0.07078 \end{aligned}$$

b) $\bar{Y} \approx N\left(2250, \frac{(10.2)^2}{15}\right)$

i.e., $\bar{Y}$ is approximately distributed as Normal with mean $= \mu_{\bar{Y}} = 2250$ and standard deviation $= \sigma_{\bar{Y}} = 2.63$

Poisson Distribution

- Poisson distribution is used to model the number of occurrences of “rare” events in certain intervals of time and space.
- Certain conditions must be met regarding the occurrence of these events and the definition of the period of time or region of space in which they occur for a Poisson random variable to represent the number of events.
- A Poisson random variable is a **discrete** random variable as it represents a *count*.

Poisson Distribution - Examples

- $X = \#$ of alpha particles emitted from a polonium bar in an 8 minute period
- $Y = \#$ of flaws on a standard size piece of manufactured product (e.g., 100m coaxial cable, 100 sq. meter plastic sheeting)
- $Z = \#$ of hits on a web page in a 24h period

Definition: The Poisson probability mass function is defined as:

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad \text{for } x = 0, 1, 2, 3, \dots$$

λ is called the *mean parameter*, and represents the *expected number* of events occurring in the interval of interest.

Computing Mean and Variance of a Poisson Random Variable

- Expected Value of $X \sim Poi(\lambda)$ is:

- $$E[X] = \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{(x-1)!} = e^{-\lambda} \lambda \sum_{x=1}^{\infty} x \frac{\lambda^{x-1}}{(x-1)!} =$$
$$e^{-\lambda} \lambda \sum_{x=0}^{\infty} x \frac{\lambda^x}{x!} = \lambda$$

- Variance of $X \sim Poi(\lambda)$ is:

- $$\text{Var}[X] = \sum_{x=0}^{\infty} (x - \lambda)^2 \left\{ \frac{e^{-\lambda} \lambda^x}{x!} \right\} = \dots = \lambda$$

Poisson distribution: Example

A manufacturer of chips produces 1% defectives. What is the probability that in a box of 100 chips no defectives are found?

Solution: A defective chip can be considered to be a rare event, since p is small ($p = 0.01$). So, model X as **Poisson** random variable.

We need to obtain a value for λ ! Look at the expected value!

Note that we expect $100 \cdot 0.01 = 1$ chip out of the box to be defective.

We know that the expected value of X is λ . In this example, therefore, we take $\lambda = 1$.

Then $P(X = 0) = \frac{e^{-1}1^0}{0!} = 0.3679$.

Poisson distribution: Example (cont.)

For the same product, what is the probability that, in a box of 200 chips, no defectives are found?

Solution: The number of defectives in a box of 200 chips Y has a Poisson distribution with parameter $\lambda = 2$ (since the expected number of defectives in 200 chips is 2.)

Then, we have $P(Y = 0) = .1353$

Note that X and Y are random variables with different Poisson distributions because the events they represent occur in different regions (two different boxes here).

Poisson to approximate Binomial

Result (not a theorem): For large n , the Binomial distribution can be approximated by the Poisson distribution, where λ is taken as $n\pi$:

$$\binom{n}{k} \pi^k (1 - \pi)^{n-k} \approx e^{-n\pi} \frac{(n\pi)^k}{k!}$$

Rule of thumb: use Poisson approximation if $n \geq 100$, $\pi \leq 0.01$ and $n\pi \leq 20$